

Thermodynamics

Lecture 3A:

- Pure substances
- Property tables

Prof. Claudio Hail (he/him)

After this lecture

- Introduction to pure substances and phase change processes.
- Illustrate (p,v) , (T,v) and (p,T) diagrams and (p,v,T) surfaces of pure substances.
- Demonstrate the procedures for determining thermodynamic properties of pure substances.

Learning objectives

What you will be able to do after this course:

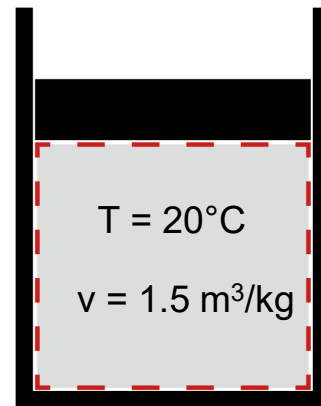
- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

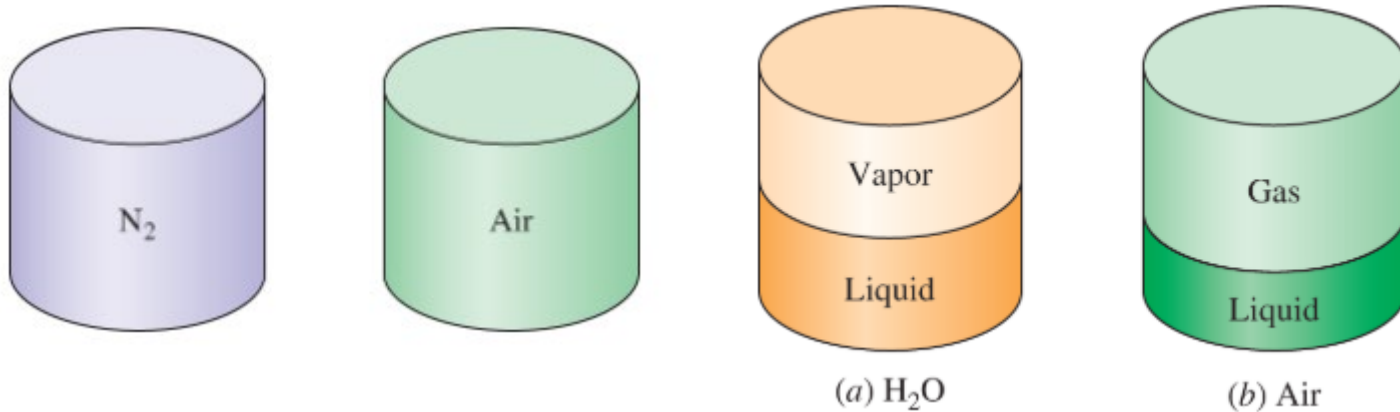
Properties of pure substances

- Thermodynamic description of the working fluids:
 - Water
 - Refrigerants (Ammonia, R134a, ...)
 - Gases – (often approximated as ideal gases)
 - Hydrocarbons
- Studying thermodynamic properties of different phases of matter.



Pure substances

- Definition: a fluid with a uniform chemical composition



Phases of matter

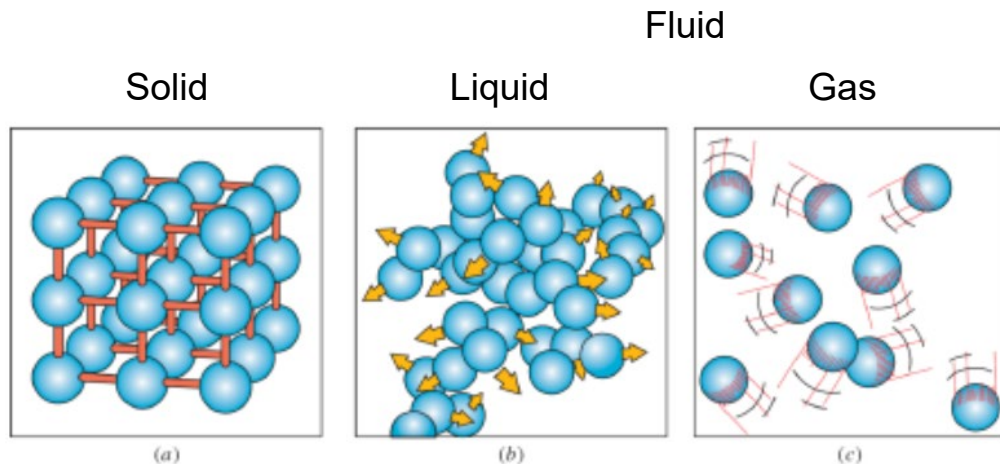


FIGURE 3–4

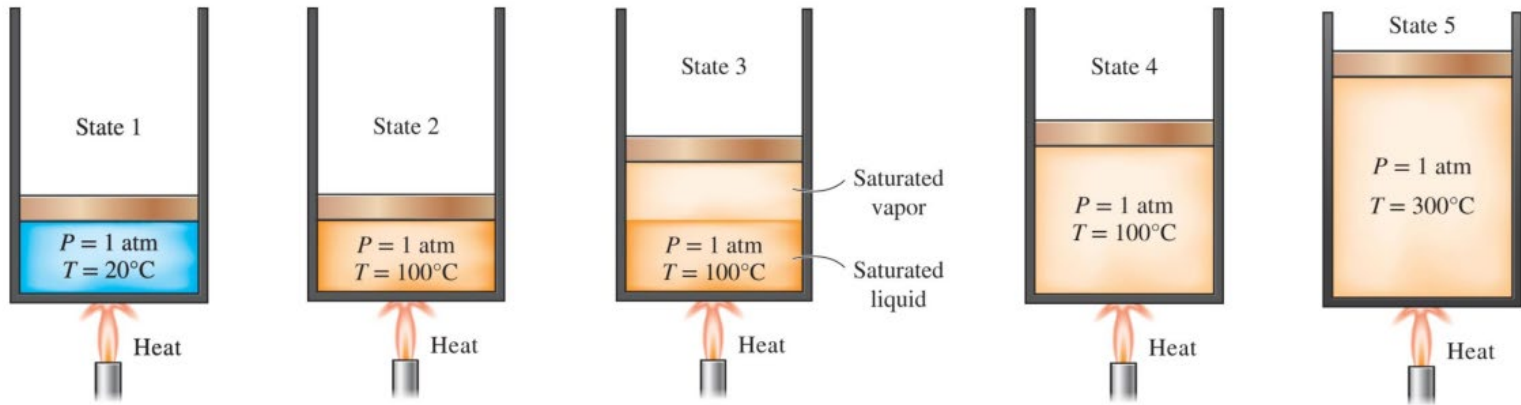
Melting
Freezing

Vaporizing
Condensation

Sublimation
Deposition (resublimation)

Phase change processes of pure substances

- Example: isobaric heating of Water



T-v diagram for pure substances



T-v diagram of a pure substance

C. Hail – ME40 – Fa26

P-v diagram for pure substances

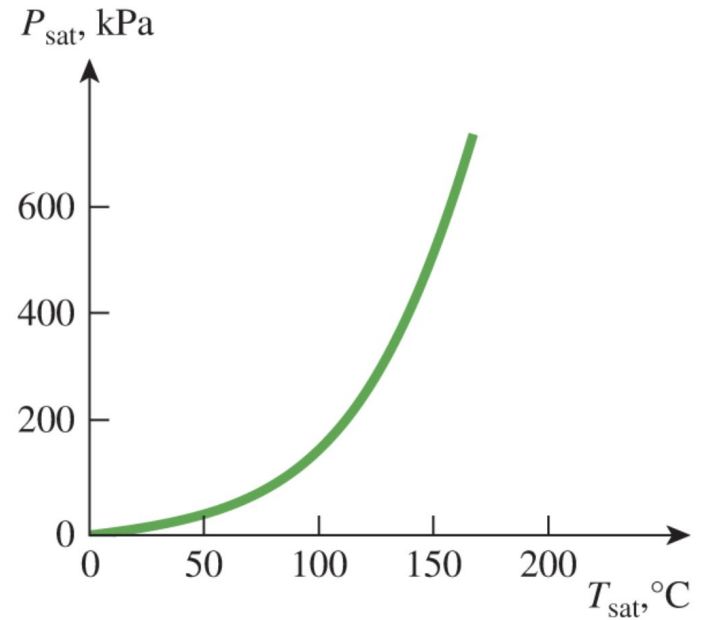


P-v diagram of a pure substance

C. Hail – ME40 –Fa26

Saturation temperature and pressure

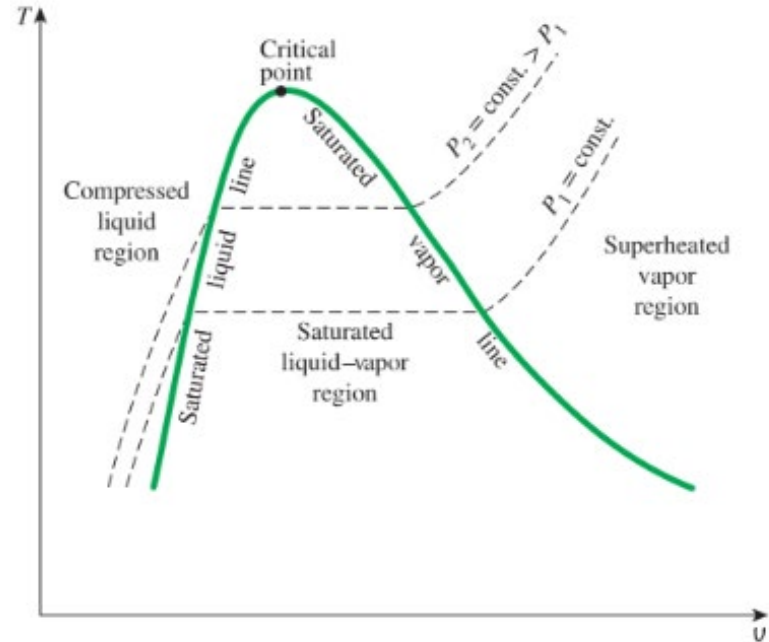
- P_{sat} pressure at which a pure substance changes phase at a given temperature.
- T_{sat} temperature at which a pure substance changes phase at a given pressure.
- During phase change process Pressure and temperature are linked.



Liquid vapor saturation curve

Critical point

- At pressures above critical point: only one phase ever, resembles vapor, but cannot tell when phase change occurred.



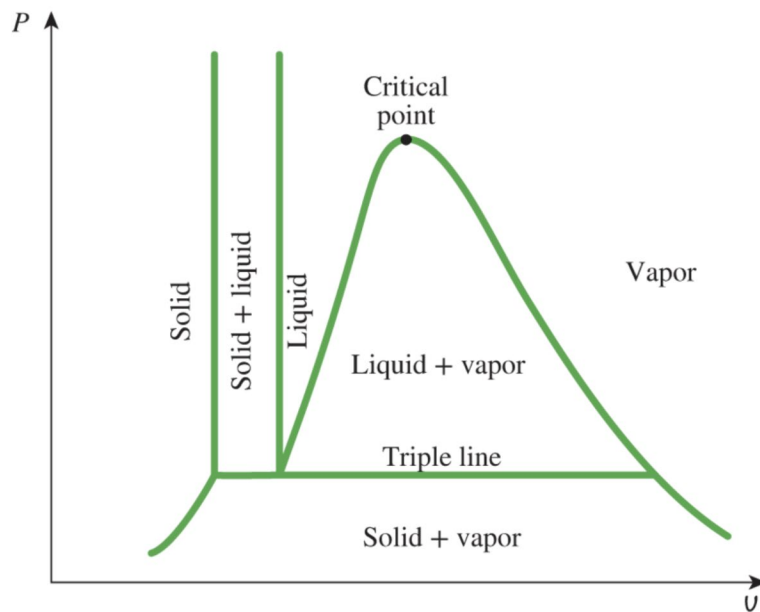
T-v diagram of a pure substance

Demonstration critical point of CO₂



Solid-Liquid-Vapor Diagram

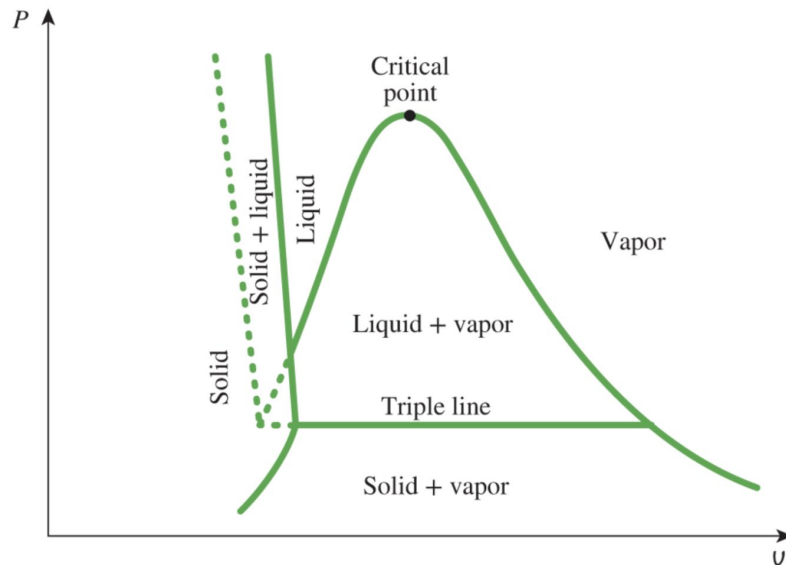
- Triple line: solid, liquid and vapor phases coexist.
- $T_{trip} = 273.16K, p_{trip} = 0.61kPa$
- Fluids can either contract or expand upon freezing.



P-v diagram of a pure substance that contracts upon freezing

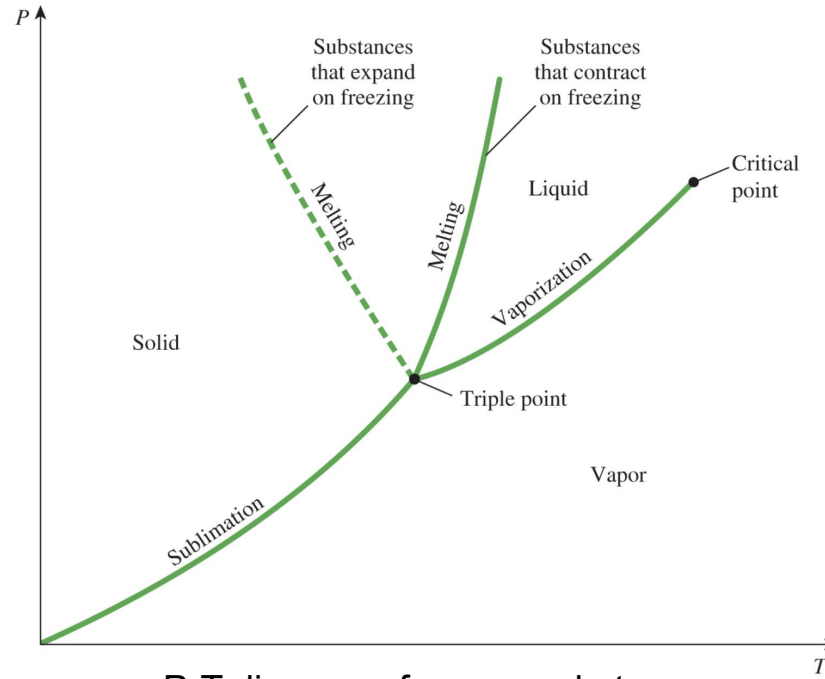
Solid-Liquid-Vapor Diagram

- Few substances expand upon freezing.
- For example, water.



P-v diagram of a pure substance that expands upon freezing

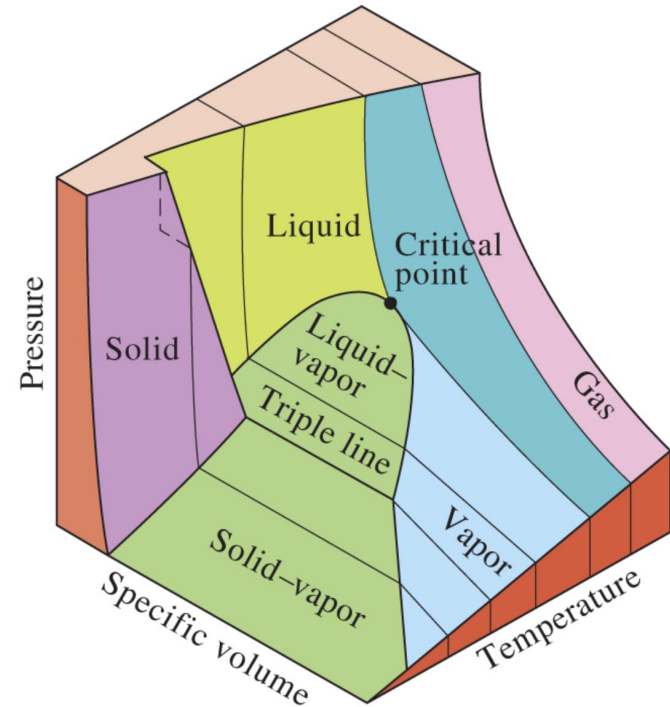
P-T diagram for pure substances



P-T diagram of a pure substance

The P-T-v surfaces in 3D

- State postulate: state of a simple compressible substance is fixed by any two independent intensive properties.



Thermodynamic property tables

Table A-1 	Molar mass, gas constant, and critical-point properties  882
Table A-2 	Ideal-gas specific heats of various common gases  883
Table A-3 	Properties of common liquids, solids, and foods  886
Table A-4 	Saturated water—Temperature table  888
Table A-5 	Saturated water—Pressure table  890
Table A-6 	Superheated water  892
Table A-7 	Compressed liquid water  896
Table A-8 	Saturated ice–water vapor  897
Figure A-9 	T-s diagram for water  898
Figure A-10 	Mollier diagram for water  899
Table A-11 	Saturated refrigerant-134a—Temperature table  900
Table A-12 	Saturated refrigerant-134a—Pressure table  902
Table A-13 	Superheated refrigerant-134a  903

Thermodynamic property tables

TABLE A-4

Saturated water—Temperature table

Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
		Sat. liquid, u_f	Sat. vapor, u_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661
25	3.1698	0.001003	43.340	104.83	2304.3	2409.1	104.83	2441.7	2546.5	0.3672	8.1895	8.5567
30	4.2469	0.001004	32.879	125.73	2290.2	2415.9	125.74	2429.8	2555.6	0.4368	8.0152	8.4520
35	5.6291	0.001006	25.205	146.63	2276.0	2422.7	146.64	2417.9	2564.6	0.5051	7.8466	8.3517
40	7.3851	0.001008	19.515	167.53	2261.9	2429.4	167.53	2406.0	2573.5	0.5724	7.6832	8.2556
45	9.5953	0.001010	15.251	188.43	2247.7	2436.1	188.44	2394.0	2582.4	0.6386	7.5247	8.1633
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898

Property tables: Saturated liquid and vapor

Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg	
		Sat. liquid, v_f	Sat. vapor, v_g
0.01	0.6117	0.001000	206.00
5	0.8725	0.001000	147.03
10	1.2281	0.001000	106.32
15	1.7057	0.001001	77.885
20	2.3392	0.001002	57.762

Subscript f: properties of saturated liquid

Subscript g: properties of saturated vapor

v_f specific volume of liquid (m^3/kg)

v_g specific volume of vapor (m^3/kg)

Property tables: Saturated liquid and vapor

Temp., T °C	Sat. Press., P_{sat} kPa	Internal energy, kJ/kg		
		Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g
0.01	0.6117	0.000	2374.9	2374.9
5	0.8725	21.019	2360.8	2381.8
10	1.2281	42.020	2346.6	2388.7
15	1.7057	62.980	2332.5	2395.5
20	2.3392	83.913	2318.4	2402.3

Subscript f: properties of saturated liquid

Subscript g: properties of saturated vapor

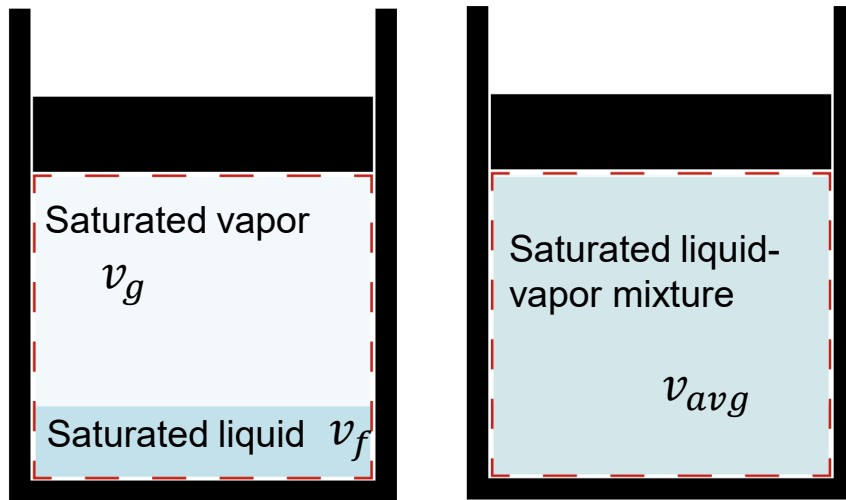
u_f specific internal energy of liquid (kJ/kg)

u_g specific internal energy of vapor (kJ/kg)

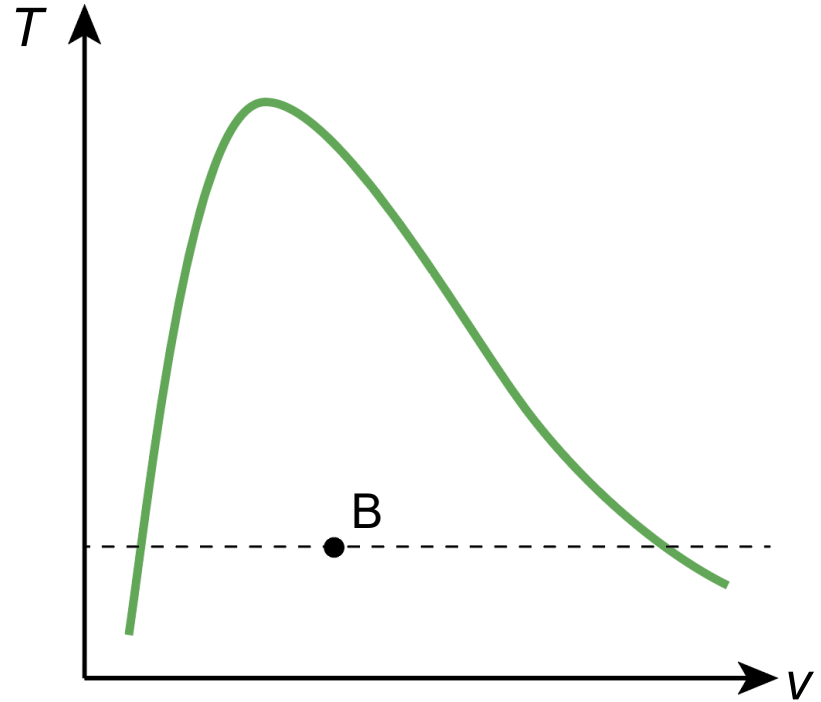
u_{fg} specific internal energy of vaporization

Property tables: Saturated liquid vapor mixture

Given: m_f, m_g determine v_{avg}



Property tables: Saturated liquid vapor mixture



Superheated vapor

How to determine $\nu(T = 50^\circ\text{C})$

How to determine $\nu(T = 75^\circ\text{C})$

T °C	ν m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$P = 0.01 \text{ MPa (} 45.81^\circ\text{C)}^*$				
Sat. †	14.670	2437.2	2583.9	8.1488
50	14.867	2443.3	2592.0	8.1741
100	17.196	2515.5	2687.5	8.4489
150	19.513	2587.9	2783.0	8.6893
200	21.826	2661.4	2879.6	8.9049
250	24.136	2736.1	2977.5	9.1015
300	26.446	2812.3	3076.7	9.2827

Excerpt from Superheated water table

Compressed liquid water

- Approximation: in the absence of compressed liquid data, treat the liquid as saturated liquid at the given temperature.

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$P = 5 \text{ MPa (263.94°C)}$				
Sat.	0.0012862	1148.1	1154.5	2.9207
0	0.0009977	0.04	5.03	0.0001
20	0.0009996	83.61	88.61	0.2954
40	0.0010057	166.92	171.95	0.5705
60	0.0010149	250.29	255.36	0.8287
80	0.0010267	333.82	338.96	1.0723
100	0.0010410	417.65	422.85	1.3034

Jupyter notebook example

Click this link to launch the notebook in the Berkeley hub: [waterProperties Script](#)

IAPWS Example

```
from iapws import IAPWS97
water = IAPWS97(P=p1_MPa, T=T1_K)
s1 = water.s # kJ/kg*K
h1 = water.h # kJ/kg
v1 = water.v # m3/kg
u1 = water.u # kJ/kg
```

CoolProp Example

```
from CoolProp.CoolProp import PropsSI
h = PropsSI('H', 'P', p1_MPa*1e6, 'T', T1_K, 'Water') # J/kg
s = PropsSI('S', 'P', p1_MPa*1e6, 'T', T1_K, 'Water') # J/(kg*K)
v = 1/PropsSI('D', 'P', p1_MPa*1e6, 'T', T1_K, 'Water') # m3/kg (D = density)
```

Summary

- Introduction to pure substances and phase change processes.
- Illustrate (p,v) , (T,v) and (p,T) diagrams and (p,v,T) surfaces of pure substances.
- Demonstrate the procedures for determining thermodynamic properties of pure substances.

Next up

- Description of ideal gasses.